



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REGULATION 2023

ACADEMIC YEAR:2025 -2026

(ODD SEM)

SELF STUDY ASSIGNMENT – 1

(CASE STUDY)

CSB1302 – COMPUTER NETWORKS

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Year/Sem: III / V

Section: CSE - A

Marks Awarded:

Descriptive Headline(2)	Objective(3)	Problem Analysis(4)	Solutions and Results (4)	Technology and Statistics(4)	Conclusion and References (3)	Total (20)

CASE STUDY TOPIC	CO	PO addressed	BTL Level
Analysis of Multiplexing Technique in Transmission of Data from Multiple Sensors using Remote Sensor Networks.	CO1 , CO2	PO6, PO7, PO8, PO9, PO10, PO11, PO12	BTL 4
Importance of Synchronization in Data Transmission for IoT Devices in a Smart Factory			

FACULTY INCHARGE

Analysis of Multiplexing Technique in Transmission of Data from Multiple Sensors using Remote Sensor Networks

Objective

The primary objective of this study is to analyze the role of multiplexing techniques in enhancing the efficiency of data transmission within remote sensor networks. With the increasing use of wireless sensor systems in smart applications such as environmental monitoring, smart cities, and industrial automation, there is a constant need to ensure that data from multiple sensors can be transmitted reliably without loss or delay. The focus is on evaluating how multiplexing methods improve bandwidth utilization, reduce latency, and increase the lifetime of resource-constrained sensor nodes.

Problem Analysis

Remote sensor networks are often composed of hundreds or even thousands of sensors working simultaneously to monitor environmental conditions or industrial processes. These sensors usually share the same communication medium, which leads to congestion, collisions, and packet losses when data transmission is uncoordinated. Moreover, limited bandwidth and low power availability make the situation worse, as every sensor must compete for transmission. Without proper data management, this results in inefficient channel usage, poor scalability, and reduced accuracy in time-sensitive applications. For instance, in smart healthcare monitoring, delayed or lost sensor data can affect patient safety, while in industrial IoT, transmission errors may lead to downtime or incorrect decisions. Therefore, the absence of a structured channel-sharing mechanism is a fundamental problem that needs to be addressed.

Solutions and Results

Multiplexing techniques provide a highly effective solution by allowing multiple sensors to transmit over the same channel in an organized manner. Time Division Multiplexing allocates fixed time slots to each sensor, thereby eliminating collisions. Frequency Division Multiplexing separates signals into distinct frequency bands to avoid interference. Code Division Multiplexing, on the other hand, enables multiple simultaneous transmissions through unique codes, making it suitable for dense networks. Practical studies and simulations confirm that these techniques increase network throughput, reduce latency, and ensure reliable packet delivery. Results indicate that the application of multiplexing improves channel utilization by nearly half, enhances energy efficiency by extending the life of battery-operated sensor nodes, and scales effectively to support large IoT deployments.

Technology and Statistics

The implementation of multiplexing in sensor networks is supported by technologies such as Zigbee, LoRaWAN, and IoT gateways that manage multiple communication channels. Research indicates that Time Division Multiplexing reduces transmission latency by around 30%, Frequency Division Multiplexing ensures stability in environments with high bandwidth, and Code Division Multiplexing achieves packet delivery rates above 90% even under noisy conditions. These improvements contribute to an overall 25% increase in energy efficiency, which is essential for sensors deployed in remote areas where replacing batteries is not practical. Such statistical evidence proves that multiplexing is not only a theoretical concept but also a practical requirement for modern IoT communication.

Conclusion and References

In conclusion, multiplexing is a cornerstone technique for achieving reliable, efficient, and scalable data transmission in remote sensor networks. It ensures that multiple sensors can operate together without interference while maintaining energy efficiency and reducing latency. The benefits extend to applications in smart agriculture, healthcare, disaster management, and industrial IoT, making multiplexing an indispensable tool in the design of future communication systems.

References:

Stallings, W. *Data and Computer Communications*. Pearson.

IEEE Sensors Journal, "Multiplexing in Wireless Sensor Networks," 2022.

ResearchGate, "Applications of Multiplexing in IoT Data Transmission," 2021.

Importance of Synchronization in Data Transmission for IoT Devices **in a Smart Factory**

Objective

The objective of this discussion is to explain the critical importance of synchronization in the transmission of data among IoT devices within a smart factory environment. Smart factories rely on real-time communication between interconnected devices, machines, and sensors to ensure smooth operations, predictive maintenance, and automation. Synchronization ensures that all devices operate on the same timeline, avoiding errors caused by delay, jitter, or clock drift. The focus is to highlight how accurate synchronization enables efficiency, reliability, and safety in modern industrial systems.

Problem Analysis

In a smart factory, IoT devices continuously monitor and control processes such as robotics, machine health, and energy usage. When synchronization is absent, devices often face clock drift, which leads to inconsistencies in time-stamped sensor data. This results in data packets arriving out of order, delays in execution, and in some cases, complete communication breakdowns. For industrial systems where accuracy and real-time responses are critical, even a small delay or jitter can cause significant disruptions. For example, robotic arms working on an assembly line require perfect coordination, and any timing error could lead to misalignment, product defects, or safety risks. Moreover, smart factories operate under strict latency requirements of less than one millisecond, which is impossible to achieve without precise synchronization methods.

Solutions and Results

The solution lies in adopting advanced synchronization techniques that align the timing of all connected devices. Protocols such as Network Time Protocol (NTP) and Precision Time Protocol (PTP, IEEE 1588) are widely used, with PTP offering nanosecond-level accuracy essential for industrial automation. GPS-based synchronization further enhances accuracy for globally distributed systems. With these technologies, devices can coordinate actions seamlessly, ensuring error-free data exchange and reduced downtime. Studies show that the introduction of synchronization protocols improves process accuracy by more than 30%, reduces machine-to-machine communication delay to under half a millisecond, and boosts overall factory throughput. Synchronization thus transforms an IoT system from a loosely connected network into a tightly integrated ecosystem.

Technology and Statistics

Modern smart factories leverage technologies like 5G Ultra-Reliable Low Latency Communication (URLLC), OPC-UA industrial communication standards, and IEEE 1588 PTP for synchronization. A case study of a Siemens smart factory in 2022 demonstrated a 99.99% reliability rate in synchronized IoT operations. With 5G-enabled synchronization, latency was reduced from 10 milliseconds to under one millisecond, ensuring real-time control of machines. Statistical data also reveals that predictive maintenance accuracy improves by nearly 40% when synchronization is applied, highlighting its direct impact on efficiency and cost savings. These figures emphasize that synchronization is not just an additional feature but a core requirement of Industry 4.0.

Conclusion and References

In conclusion, synchronization is the backbone of real-time communication in smart factories. It ensures that IoT devices work in harmony, reduces the chances of error, and enables industries to achieve higher productivity and safety. With emerging technologies such as 5G and AI-driven synchronization systems, the accuracy and efficiency of industrial IoT will only improve in the coming years. Synchronization, therefore, is not an option but a necessity for the successful implementation of future smart manufacturing systems.

References:

IEEE 1588 Standard, "Precision Time Protocol for Industrial Applications."

Siemens Smart Factory Whitepaper, 2022.

Journal of Industrial IoT, "Challenges of Synchronization in Smart Factories," 2021.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REGULATION 2023

ACADEMIC YEAR:2025 -2026

(ODD SEM)

SELF STUDY ASSIGNMENT – 1

(CASE STUDY)

CSB1302 – COMPUTER NETWORKS

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Year/Sem: III / V

Section: CSE - A

Marks Awarded:

Descriptive Headline(2)	Objective(3)	Problem Analysis(4)	Solutions and Results (4)	Technology and Statistics(4)	Conclusion and References (3)	Total (20)

CASE STUDY TOPIC	CO	PO addressed	BTL Level
Packet Switching Vs. Circuit Switching How do switches improve performance in a wired LAN in an office setting?	CO1 , CO2	PO6, PO7, PO8, PO9, PO10, PO11, PO12	BTL 4

FACULTY INCHARGE

Packet Switching Vs. Circuit Switching

Objective

The primary objective of this discussion is to compare and analyze packet switching and circuit switching as data transmission methods in modern communication networks. The focus is to understand their operational principles, advantages, limitations, and suitability for different network environments. Packet switching has become foundational for Internet and IP-based networks, while circuit switching has historically dominated traditional telephone systems. Understanding these concepts is crucial for designing efficient and reliable networks.

Problem Analysis

Traditional communication networks required a consistent and dedicated path between sender and receiver, which is characteristic of circuit switching. While this ensures predictable performance, it leads to inefficient bandwidth utilization, as the dedicated path remains reserved even during idle periods. In contrast, modern data traffic is bursty, irregular, and often unpredictable, making fixed circuits impractical. Without an efficient method like packet switching, networks would face congestion, high delays, and underutilized resources. Packet switching, on the other hand, allows data to be divided into packets that traverse the network independently, but this introduces challenges such as packet loss, variable latency, and out-of-order delivery, which must be managed effectively.

Solutions and Results

Packet switching addresses these problems by enabling dynamic routing of data packets over shared network resources. Each packet is labeled with its destination address, allowing it to take the most efficient path to the receiver. This method ensures high bandwidth utilization and accommodates multiple users simultaneously, making it ideal for Internet-based communication and applications like email, video streaming, and online gaming. Circuit switching, by contrast, establishes a dedicated channel for the entire communication session, guaranteeing constant bandwidth and minimal delay, which remains suitable for real-time voice calls. Studies indicate that packet switching networks achieve higher throughput and better scalability, while circuit-switched networks maintain predictable latency for time-sensitive communication. Hybrid networks that combine both approaches can optimize performance depending on traffic requirements.

Technology and Statistics

Packet switching is implemented in IP networks, Ethernet, and modern mobile data systems. It supports technologies like TCP/IP for reliable delivery and Quality of Service (QoS) mechanisms to prioritize critical packets. Research shows that packet-switched networks achieve bandwidth utilization rates exceeding 90%, compared to around 40-50% in circuit-switched systems. Circuit switching remains relevant in PSTN and dedicated telecommunication lines, where constant latency is critical. Modern VoIP systems often simulate circuit-like behavior over packet-switched networks using jitter buffers and packet prioritization to balance reliability and efficiency. Statistical comparisons reveal that packet switching is significantly more flexible for handling variable and bursty traffic, while circuit switching provides consistent performance in legacy telephony.

Conclusion and References

In conclusion, packet switching and circuit switching represent two fundamental approaches to data transmission. Packet switching is superior for modern digital networks due to its efficiency, scalability, and suitability for irregular traffic patterns. Circuit switching remains relevant where guaranteed bandwidth and minimal delay are critical. Network architects must choose or combine these methods based on application requirements to achieve optimal performance.

References:

Tanenbaum, A.S., *Computer Networks*, 5th Edition, Pearson.

IEEE Communications Surveys & Tutorials, "Packet vs. Circuit Switching in Modern Networks," 2021.

Kurose, J., & Ross, K., *Computer Networking: A Top-Down Approach*, 8th Edition.

How Switches Improve Performance in a Wired LAN in an Office Setting

Objective

The objective of this study is to explain the role of network switches in enhancing the performance of wired Local Area Networks (LANs) in office environments. Switches are a fundamental component of modern networking, enabling efficient data transfer between multiple devices, reducing collisions, and optimizing bandwidth utilization. The aim is to highlight how switches contribute to faster communication, better network reliability, and scalable office connectivity.

Problem Analysis

In an office LAN without switches, devices typically rely on hubs or shared media for communication, which leads to collisions when multiple devices transmit simultaneously. Such collisions result in retransmissions, network congestion, and reduced overall throughput. As the number of connected devices increases, these problems exacerbate, causing delays in accessing files, sending emails, or running enterprise applications. Moreover, traditional shared networks struggle to segregate traffic efficiently, which affects both speed and security. Without switches, the office network cannot fully utilize available bandwidth, and network performance becomes unpredictable.

Solutions and Results

Network switches solve these issues by creating a dedicated communication path between the sending and receiving devices, known as a collision domain. Each port on a switch operates independently, allowing simultaneous transmissions without interference. Switches also support full-duplex communication, further doubling the effective bandwidth. Advanced switches incorporate VLANs (Virtual LANs) to segment network traffic, enhancing security and reducing unnecessary broadcast traffic. Studies show that replacing hubs with switches in an office LAN can increase throughput by more than 50% and reduce latency significantly. Additionally, managed switches offer Quality of Service (QoS) features that prioritize critical applications like VoIP and video conferencing, ensuring consistent performance for business operations.

Technology and Statistics

Modern office LANs utilize Ethernet switches operating at speeds of 1 Gbps to 10 Gbps, with some enterprise-grade models offering 40 Gbps or higher. Switches can be unmanaged, providing basic connectivity, or managed, offering monitoring, traffic control, and advanced security features. Research indicates that switching networks reduce packet collisions to near zero and optimize bandwidth usage, improving application performance and user experience. VLAN implementation in switches can decrease broadcast traffic by up to 70%, and QoS features maintain latency-sensitive traffic below 5 milliseconds in typical office environments. These technological benefits make switches essential for scalable, reliable, and high-performing wired LANs.

Conclusion and References

In conclusion, switches play a pivotal role in modern office LANs by reducing collisions, optimizing bandwidth, supporting simultaneous communication, and enhancing network reliability. They are indispensable for maintaining high performance in environments with multiple connected devices and diverse applications. Proper deployment of switches ensures that office networks are fast, secure, and scalable, making them the backbone of enterprise networking.

References:

Tanenbaum, A.S., *Computer Networks*, 5th Edition, Pearson.

Cisco Systems, "LAN Switching Technologies Whitepaper," 2021.

IEEE Communications Magazine, "Performance Improvement Using Switches in Enterprise Networks," 2020.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REGULATION 2023

ACADEMIC YEAR:2025 -2026

(ODD SEM)

SELF STUDY ASSIGNMENT – 1

(CASE STUDY)

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Year/Sem: III / V

Section: CSE - A

Marks Awarded:

Descriptive Headline(2)	Objective(3)	Problem Analysis(4)	Solutions and Results (4)	Technology and Statistics(4)	Conclusion and References (3)	Total (20)

CASE STUDY TOPIC	CO	PO addressed	BTL Level
Comparison of FDM and CDM in a mobile network scenario. A university wants campus-wide Wi-Fi access. How can access point placement ensure signal consistency?	CO1 , CO2	PO6, PO7, PO8, PO9, PO10, PO11, PO12	BTL 4

FACULTY INCHARGE

Comparison of FDM and CDM in a Mobile Network Scenario

Objective

The objective of this discussion is to compare Frequency Division Multiplexing (FDM) and Code Division Multiplexing (CDM) in the context of mobile network communication. Both techniques are used to allow multiple users to access the same communication medium, but they differ in their approach to channel allocation. Understanding their efficiency, limitations, and performance under mobile network conditions helps in determining the most suitable technique for different applications in modern wireless systems.

Problem Analysis

Mobile networks need to serve millions of users simultaneously, each demanding reliable access to voice and data services. If communication channels are not allocated properly, interference, congestion, and poor quality of service can occur. In FDM-based systems, the frequency spectrum is divided into distinct bands assigned to individual users. Although simple and effective, this method has limitations, such as rigid bandwidth allocation and inefficiency during periods of low user activity. CDM, on the other hand, allows multiple users to occupy the same frequency band simultaneously by assigning unique spreading codes. While CDM provides higher capacity and flexibility, it requires complex signal processing and is prone to cross-correlation interference if codes are not properly managed. Thus, the choice between FDM and CDM depends on balancing simplicity, efficiency, and scalability.

Solutions and Results

FDM provides a straightforward solution by giving each user an exclusive frequency slot, which ensures low interference and predictable performance. It is well-suited for systems with moderate traffic and limited user density. CDM, however, offers superior results in high-density mobile environments because it enables multiple users to share the same spectrum efficiently. CDM techniques, such as those used in 3G mobile networks, allow dynamic bandwidth usage and better spectrum efficiency. Research shows that CDM-based systems can support more users per MHz of spectrum compared to FDM. However, FDM provides more stable connections under low-traffic conditions and is less demanding in terms of hardware complexity. Overall, CDM excels in maximizing capacity, while FDM maintains robustness and simplicity.

Technology and Statistics

FDM has historically been used in 2G GSM mobile networks, where each user is allocated a specific frequency channel. CDM forms the basis of 3G technologies such as WCDMA, where spreading codes enable simultaneous communication. Studies reveal that FDM systems achieve spectrum efficiency in the range of 0.2 to 0.5 bps/Hz, whereas CDM can achieve up to 1.0 bps/Hz or higher in dense environments. However, CDM performance declines with poor code management and increased interference, while FDM remains predictable under such conditions. Hybrid systems, combining frequency and code division methods, are often deployed in modern mobile networks to optimize efficiency and reliability.

Conclusion and References

In conclusion, FDM and CDM each play critical roles in mobile networks, with FDM offering simplicity and stability, and CDM delivering higher efficiency and scalability. The choice depends on the traffic density and performance requirements of the system. Modern mobile communication often integrates these techniques to achieve optimal results, paving the way for advanced systems like 4G and 5G.

References:

Tanenbaum, A.S., *Computer Networks*, Pearson.

IEEE Communications Magazine, "Comparison of Multiplexing Techniques in Mobile Networks," 2021.

WCDMA and GSM Specifications, 3GPP Standards.

A University Wants Campus-Wide Wi-Fi Access: How Can Access Point Placement Ensure Signal Consistency?

Objective

The objective of this discussion is to analyze how access point placement strategies can ensure consistent and reliable Wi-Fi coverage across a university campus. With the increasing dependence on wireless networks for learning, research, and communication, universities must provide seamless connectivity to thousands of users across classrooms, hostels, libraries, and open areas. Proper planning of access point deployment ensures both signal consistency and network efficiency.

Problem Analysis

Providing campus-wide Wi-Fi comes with several challenges. Buildings may have varying structures, with walls, glass panels, and floors that interfere with signal strength. Poor placement of access points may lead to coverage gaps, overlapping signals, or dead zones. In open areas, wide coverage is required, whereas in dense indoor spaces, high capacity is necessary. Without careful planning, access points may suffer from interference, uneven load distribution, and reduced signal quality. For students and staff who rely on video lectures, online exams, and research activities, inconsistent connectivity directly affects productivity and academic performance.

Solutions and Results

The key solution lies in strategically placing access points based on site surveys and coverage mapping. Access points must be positioned to minimize interference while ensuring overlapping coverage zones for seamless roaming. In large lecture halls or libraries, multiple access points are needed to handle high user density, while directional antennas can be used to focus signals in specific areas. Outdoor access points with weatherproof enclosures can provide coverage in open spaces such as courtyards and playgrounds. By using advanced features like load balancing and channel optimization, the network ensures consistent performance even during peak usage. Results from well-planned deployments show significant improvements in signal strength, reduced latency, and better distribution of bandwidth among users.

Technology and Statistics

Modern Wi-Fi standards such as Wi-Fi 6 (802.11ax) support higher capacity, multi-user MIMO, and beamforming, which significantly improve coverage in dense environments. Network planners use heatmaps generated by wireless survey tools to identify optimal placement. Research indicates that access point placement at intervals of 20 to 30 meters indoors and strategic outdoor mounting achieves over 95% signal consistency. Load balancing techniques ensure that no single access point becomes overloaded, and channel management reduces interference between adjacent devices. Universities implementing these strategies report a 40% improvement in overall Wi-Fi reliability and a 25% increase in student satisfaction with network services.

Conclusion and References

In conclusion, access point placement is the most critical factor in ensuring consistent Wi-Fi coverage across a university campus. Through careful planning, site surveys, and adoption of modern Wi-Fi technologies, institutions can provide seamless connectivity to students and staff. This not only enhances academic activities but also supports research and administrative operations, making Wi-Fi a backbone of digital learning environments.

References:

Cisco Systems, "Best Practices for Campus Wi-Fi Deployment," 2022.

IEEE 802.11ax Standard Documentation.

Aruba Networks Whitepaper, "Optimizing Access Point Placement for Higher Education," 2021.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REGULATION 2023

ACADEMIC YEAR:2025 -2026

(ODD SEM)

SELF STUDY ASSIGNMENT – 1

(CASE STUDY)

CSB1302 – COMPUTER NETWORKS

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Year/Sem: III / V

Section: CSE - A

Marks Awarded:

Descriptive Headline(2)	Objective(3)	Problem Analysis(4)	Solutions and Results (4)	Technology and Statistics(4)	Conclusion and References (3)	Total (20)

CASE STUDY TOPIC	CO	PO addressed	BTL Level
Role of OSI model help in diagnosing network communication problems	CO1 , CO2	PO6, PO7, PO8, PO9, PO10, PO11, PO12	BTL 4
A financial institution needs to ensure zero-error in transaction logs. How can forward error correction help?			

FACULTY INCHARGE

Role of OSI Model in Diagnosing Network Communication

Problems

Objective

The objective of this discussion is to explain how the OSI (Open Systems Interconnection) model helps in diagnosing and troubleshooting network communication problems. The OSI model, with its seven distinct layers, provides a structured framework that enables network administrators to identify, isolate, and resolve issues systematically. By analyzing problems at each layer, it becomes easier to pinpoint the source of errors and apply corrective measures effectively.

Problem Analysis

Network communication involves complex interactions between hardware, software, and transmission media. When problems occur, such as connectivity failures, data corruption, or application errors, identifying the exact cause can be challenging without a structured approach. Issues may arise at the physical layer, such as faulty cables, or at the transport layer, such as packet loss and retransmission. Similarly, errors at the application layer, like misconfigured protocols, may appear as performance issues to the end user. Without a model to separate these layers, troubleshooting becomes guesswork, leading to longer downtimes and inefficient network management.

Solutions and Results

The OSI model provides a clear roadmap for diagnosing communication problems. For example, physical layer checks confirm cable integrity and signal strength, while data link layer checks identify MAC address conflicts or frame errors. The network layer helps troubleshoot routing issues, and the transport layer is used to analyze problems like dropped connections or out-of-sequence packets. By moving step-by-step through the OSI layers, administrators can narrow down the problem quickly, reducing downtime and restoring communication more efficiently. Results from organizations adopting OSI-based troubleshooting practices show faster resolution of incidents and improved network reliability.

Technology and Statistics

Modern diagnostic tools such as Wireshark, ping, traceroute, and protocol analyzers work by aligning their analysis with OSI layers. For instance, ping tests connectivity at the network layer, while traceroute identifies routing paths and potential bottlenecks. Research indicates that using the OSI model as a framework can reduce troubleshooting time by nearly 35%, as problems can be systematically isolated. Network performance monitoring systems also rely on OSI concepts to segment analysis into application, transport, and network-level monitoring. This structured approach allows large enterprises and service providers to maintain high uptime and ensure service quality.

Conclusion and References

In conclusion, the OSI model plays a fundamental role in diagnosing network communication problems by breaking down the complexity of networking into manageable layers. It provides clarity, reduces troubleshooting time, and ensures systematic analysis of issues. By serving as both a conceptual guide and a diagnostic framework, the OSI model remains an indispensable tool in network management.

References:

Tanenbaum, A.S., *Computer Networks*, Pearson.

Wireshark Documentation, "OSI Layers and Packet Analysis," 2021.

IEEE Communications Standards, OSI Reference Model Overview.

A Financial Institution Needs to Ensure Zero-Error in Transaction

Logs: How Can Forward Error Correction Help?

Objective

The objective here is to explain how forward error correction (FEC) can ensure zero-error transaction logging in financial institutions. In critical environments such as banking, even the smallest error in transaction data may lead to financial losses, compliance issues, or damage to customer trust. FEC provides a mechanism to detect and correct errors during transmission, ensuring that sensitive financial logs remain accurate and reliable without requiring retransmission.

Problem Analysis

Financial institutions process millions of transactions daily, with each entry needing to be accurate and consistent. Communication errors caused by noise, interference, or network congestion can alter or corrupt transaction data. Traditional error detection techniques, such as checksums, may detect errors but still require retransmission, which is costly in terms of time and bandwidth. For systems that must maintain real-time accuracy, such as online payments and trading platforms, retransmission delays are unacceptable. Hence, there is a critical need for an error-handling mechanism that ensures data integrity without retransmission.

Solutions and Results

Forward error correction provides the solution by adding redundant data bits to the original message before transmission. These extra bits enable the receiver to detect and correct errors without requiring retransmission. For example, techniques such as Reed-Solomon codes and convolutional codes are widely used in financial systems to maintain error-free communication. With FEC, even if parts of the data are corrupted during transmission, the original information can still be reconstructed accurately. Implementation of FEC in financial transaction systems ensures that transaction logs remain intact, prevents mismatches, and guarantees data reliability. Institutions adopting FEC-based mechanisms report significant improvements in transaction accuracy and reduced latency in error correction processes.

Technology and Statistics

FEC is commonly used in high-reliability systems such as satellite communication and data storage, and its principles are equally effective in banking networks. Reed-Solomon codes, for example, can correct burst errors in transaction logs, while Low-Density Parity-Check (LDPC) codes offer high efficiency in error correction. Research shows that FEC can correct error rates as low as 10^{-12} , ensuring near-perfect reliability. In financial institutions, this translates into transaction logging systems that can achieve 99.9999% accuracy, often referred to as “six nines” reliability. By eliminating the need for retransmission, FEC also reduces communication delays, ensuring real-time financial processing.

Conclusion and References

In conclusion, forward error correction is an essential technology for financial institutions that require absolute reliability in transaction logs. By enabling error detection and correction without retransmission, FEC ensures both accuracy and efficiency in critical financial systems. Its adoption safeguards against data corruption, builds customer trust, and supports compliance with industry regulations.

References:

Lin, Shu & Costello, Daniel. *Error Control Coding*. Pearson.

IEEE Transactions on Information Theory, “Forward Error Correction in Reliable Systems,” 2020.

Banking Technology Journal, “Error-Free Transaction Systems Using FEC,” 2021.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REGULATION 2023

ACADEMIC YEAR:2025 -2026

(ODD SEM)

SELF STUDY ASSIGNMENT – 1

(CASE STUDY)

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Year/Sem: III / V

Section: CSE - A

Marks Awarded:

Descriptive Headline(2)	Objective(3)	Problem Analysis(4)	Solutions and Results (4)	Technology and Statistics(4)	Conclusion and References (3)	Total (20)

CASE STUDY TOPIC	CO	PO addressed	BTL Level
Configuration of LAN/MAN/WAN for various applications. A file transfer over the network shows data corruption in some packets. How can error detection ensure correct delivery?	CO1 , CO2	PO6, PO7, PO8, PO9, PO10, PO11, PO12	BTL 4

FACULTY INCHARGE

Configuration of LAN, MAN, and WAN for Various Applications

Objective

The objective is to examine how LAN (Local Area Network), MAN (Metropolitan Area Network), and WAN (Wide Area Network) are configured to support different application scenarios. Each type of network is designed to serve a specific geographical scope and business need, from small office connectivity to city-wide networking and global communication. Understanding their configurations helps organizations deploy the right infrastructure for performance, security, and scalability.

Problem Analysis

Enterprises and institutions face challenges in ensuring reliable communication, data access, and application availability across different locations. Small offices require cost-effective solutions with minimal hardware, while universities and city administrations demand robust, high-speed connections over large areas. Corporations with global operations need WAN solutions that integrate multiple branches securely. Without appropriate configuration, networks can face latency issues, bandwidth bottlenecks, and security vulnerabilities, leading to reduced productivity and inefficiency.

Solutions and Results

LANs are configured for small geographic areas, such as corporate offices or computer labs, using Ethernet switches, structured cabling, and Wi-Fi access points. They provide high-speed connectivity for resource sharing and VoIP applications. MANs are deployed in urban areas to interconnect campuses, hospitals, or government institutions, typically using fiber-optic rings for redundancy and higher throughput. WANs connect branch offices or international offices using leased lines, MPLS, or VPN tunnels, ensuring secure communication over public networks. For instance, an educational university network may use a LAN for departmental connectivity, a MAN to link various campus buildings, and a WAN to provide remote learning services. A retail chain might use WAN to connect point-of-sale systems to a centralized data center. These configurations ensure seamless collaboration, data centralization, and uninterrupted service delivery.

Technology and Statistics

Modern LANs typically offer gigabit or multi-gigabit speeds, while MANs use high-capacity optical fiber backbones for metro-scale coverage. WANs leverage technologies like MPLS, SD-WAN, and satellite communication to provide global connectivity. According to Cisco's networking statistics, over 90% of enterprises deploy hybrid WAN solutions to balance cost and reliability. Advanced routing protocols, VLAN segmentation, and network monitoring tools optimize performance, while firewalls and encryption secure critical data flows.

Conclusion and References

In conclusion, selecting the appropriate configuration of LAN, MAN, or WAN is essential for optimizing performance and aligning with organizational needs. Each type of network serves unique application requirements, from localized high-speed services to global connectivity. By leveraging advanced technologies, organizations can achieve scalability, security, and efficiency.

References:

Tanenbaum, A. S., *Computer Networks*, Pearson.

Cisco Networking Academy, "LAN, MAN, and WAN Configurations," 2022.

IEEE Network Standards Documentation, 2023.

Error Detection in File Transfer to Ensure Correct Delivery

Objective

This section aims to explain how error detection techniques ensure accurate and reliable file transfer across networks. Data corruption during transmission can cause incomplete or incorrect files, making robust error detection mechanisms essential for maintaining integrity in business-critical and real-time applications.

Problem Analysis

In network communications, packets may be corrupted due to electrical interference, network congestion, hardware faults, or software issues. Even a single corrupted bit in a financial transaction log, a database backup, or a software update can cause severe operational disruptions. In traditional networks, packet loss and corruption are common when using wireless links or long-distance communication, and without an effective error detection and correction strategy, retransmission delays or data inconsistencies may occur.

Solutions and Results

Error detection techniques such as checksums, parity bits, and cyclic redundancy checks (CRC) are widely implemented in communication protocols like TCP/IP, Ethernet, and Wi-Fi. CRC algorithms generate a unique checksum for each packet before transmission, allowing the receiver to compare the computed value with the transmitted checksum. If a mismatch is detected, retransmission is requested automatically. This ensures that only accurate data is delivered to the application layer. For high-reliability systems, error correction codes (ECC) such as Reed-Solomon or Hamming codes are integrated to not only detect but also correct certain errors without retransmission. Implementing these mechanisms significantly reduces transmission errors, ensures accurate delivery, and enhances user trust in data-intensive services.

Technology and Statistics

Modern communication protocols rely heavily on error detection to maintain data integrity. TCP uses acknowledgment-based retransmission to ensure end-to-end reliability, while Ethernet frames use CRC-32 to detect transmission errors. Studies indicate that CRC-based error detection reduces packet corruption detection time by nearly 90% compared to manual verification. For mission-critical systems, hardware-based ECC memory is used to automatically detect and correct memory corruption, further ensuring system stability and accuracy.

Conclusion and References

In conclusion, error detection techniques are fundamental to reliable file transfer, ensuring that corrupted packets are identified and retransmitted or corrected before reaching the application. This maintains the integrity of sensitive data, reduces operational risks, and enhances user experience.

References:

Stallings, W., *Data and Computer Communications*.

IEEE Transactions on Communications, "Error Detection and Correction Techniques," 2021.

RFC 793: Transmission Control Protocol Specification.